

CO4_AT1 – DATA INTERPRETATION

Revised Answers with Additional Explanation

1. Real-Time Detection System

A surveillance system must detect objects continuously while responding quickly. The supplied data compares Viola-Jones, YOLO and Deep Learning in terms of accuracy and speed.

Method	Accuracy	Speed
Viola-Jones	78%	Fast
YOLO	92%	Moderate
Deep Learning	95%	Slow

Evaluation: Viola-Jones has the fastest response but its 78% accuracy is comparatively low. Deep Learning provides the highest accuracy of 95%, but its slow processing conflicts with strict latency requirements. YOLO achieves 92% accuracy with moderate speed, giving a better balance between detection quality and response time.

Additional interpretation: In a practical surveillance environment, an algorithm should not be selected using accuracy alone. Response time, available computing resources and the consequence of missed detections should also be considered. If the latency requirement is extremely strict, Viola-Jones may be acceptable. If accuracy and speed must both be considered, YOLO is more suitable.

Final Answer: YOLO is the recommended method because it provides 92% accuracy with moderate speed and offers the best practical compromise for the stated condition.

Conclusion: The choice depends on deployment requirements. YOLO is preferred when acceptable accuracy and timely response are required together.

2. Robust Object Detection

Robustness refers to how well a detection method maintains its performance when the visual environment becomes difficult, such as during occlusion or low-light conditions.

Method	Normal	Occlusion
Viola-Jones	80%	60%
YOLO	90%	75%
Deep Learning	95%	85%

Low-Light Results: Viola-Jones = 55%, YOLO = 70%, Deep Learning = 80%.

Evaluation: Deep Learning performs best under all three conditions. It reaches 95% in normal scenes, 85% under occlusion and 80% in low light. YOLO is second with 90%, 75% and 70%, while Viola-Jones records the lowest results.

Additional interpretation: The decline in accuracy shows that environmental conditions affect all methods. However, Deep Learning loses less performance as the scene becomes difficult, indicating stronger robustness. The trade-off is higher computational demand, whereas YOLO offers a useful middle ground for systems where resources or response time are limited.

Final Answer: Deep Learning is the best overall method for robust object detection because it achieves the highest accuracy in normal, occluded and low-light conditions.

Conclusion: Deep Learning is preferred for maximum robustness, YOLO for balance, and Viola-Jones for less demanding environments.

3. Optical Flow Analysis

The question describes two optical-flow methods without numerical accuracy values. Therefore, the comparison is based on stability, fine-motion detection, noise sensitivity and application requirements.

Criterion	Method A	Method B
Motion stability	High	Lower
Fine-motion detection	Lower	High
Noise sensitivity	Lower	Higher

Method A: Produces stable and consistent motion vectors. It is useful for tracking large or dominant movement, but it may miss small motions.

Method B: Detects fine and subtle movements, but its vectors are noisy. Noise can cause fluctuating directions and reduce tracking stability.

Additional interpretation: For a general dynamic-scene tracking system, stability is important because the estimated trajectory should not fluctuate unnecessarily. Therefore, Method A is the safer choice when smooth and reliable tracking is the main objective. Method B becomes preferable when detecting very small movements is more important and noise can be filtered.

Final Answer: Method A is recommended for general dynamic motion tracking because its stable vectors provide consistent tracking. Method B is suitable for fine-motion analysis when small movements must be detected.

Conclusion: The final selection should depend on whether the application values stable tracking or fine-motion sensitivity more strongly.

4. Model Generalization and Overfitting

The supplied data compares training and testing accuracy. The difference between these values helps identify overfitting and generalization ability.

Model	Training Accuracy	Testing Accuracy
A	98%	70%
B	90%	88%

Model A: The training accuracy is 98%, but testing accuracy drops to 70%. The train-test gap is 28 percentage points. This large gap indicates a high risk of overfitting. The model performs very well on training data but does not transfer its performance effectively to unseen data.

Model B: Training accuracy is 90% and testing accuracy is 88%, giving only a 2 percentage-point gap. This indicates stronger generalization and lower overfitting risk.

Additional interpretation: For real-world deployment, testing performance is more informative than training performance alone because actual inputs may differ from the examples used for training. A model with a small accuracy gap is generally more consistent on unseen data.

Final Answer: Model B should be selected for real-world deployment because it has 88% testing accuracy and only a 2-point train-test gap, indicating better generalization and lower overfitting risk.

Conclusion: Model B is more reliable despite having lower training accuracy because it performs consistently on unseen data.

5. System Performance under Constraints

For real-time applications, accuracy must be considered together with latency and cost. The supplied data shows a clear trade-off among these three factors.

Model	Accuracy	Latency
A	88%	50 ms
B	92%	80 ms
C	95%	120 ms

Cost: Model A = Low, Model B = Moderate, Model C = High.

Model A: It has the lowest latency (50 ms) and low cost, making it attractive when rapid response and budget are the main concerns. Its limitation is the lowest accuracy of 88%.

Model B: It provides 92% accuracy, 80 ms latency and moderate cost. Compared with Model A, it gains 4 percentage points of accuracy while adding 30 ms of latency. It therefore represents a balanced middle option.

Model C: It provides the highest accuracy at 95%, but its 120 ms latency and high cost make it less attractive for applications where real-time response and affordability are important.

Additional interpretation: Selecting the highest-accuracy model automatically would ignore deployment constraints. A practical system should consider whether the improvement in accuracy justifies additional delay and cost. Under the general requirement of balancing all three factors, Model B gives the most reasonable compromise.

Final Answer: Model B is the best choice for the stated real-time application because it offers 92% accuracy, 80 ms latency and moderate cost.

Conclusion: Model A is best for minimum latency and low cost, Model C is best for maximum accuracy, and Model B is the best overall balance.